

Exam — Software Construction 2

1. Notice

- Please write your name and matriculation number on the top of the paper.
- You may use a cheat sheet of either 2 double-sided A4 paper or 4 one-sided A4 paper. No sticky notes allowed!
- You have 120 minutes of time for the exam.
- If you have further questions, please take it up to the staff.

2. Theoretical Part: 60min

Task A — Basic Concepts

10 P.

Explain the following five concepts from object-oriented programming and provide an example for each:

- *Class*
- *Object*
- *Inheritance*
- *Composition*
- *Method Overriding*

Task B — Concepts of Object-Oriented Programming

15 P.

Briefly describe what the various terms and concepts mean in Java or object-oriented programming.

1. Define the term “polymorphism” and provide an example for each type in Java.

5 P.

2. Explain what static methods and static attributes are in Java. How do they differ from instance methods and attributes? Provide an example. 5 P.

3. Explain the differences between an abstract class and an interface in Java. When would you use one over the other? Provide an example for each. 5 P.

Task C – True or False

16 P.

Decide whether the following statements are true or false:

Question	True	False
<i>In Java, a class can inherit from multiple other classes.</i>		
<i>Static methods can be overridden in subclasses.</i>		
<i>A constructor must have the same name as the class.</i>		
<i>Private methods are inherited by subclasses but cannot be accessed.</i>		
<i>An interface can contain concrete (implemented) methods.</i>		
<i>The this keyword refers to the current instance of a class.</i>		
<i>Abstract classes can be instantiated directly.</i>		
<i>Method overloading is an example of compile-time polymorphism.</i>		

Task D — Code Completion

12 P.

Complete the following `Rectangle` class by filling in the missing code directly in the gaps. The class should represent a rectangle with width and height.

```
1  public class Rectangle {
2      // TODO: Add private attributes for width and height (both double)
3
4
5      // TODO: Create a constructor that takes width and height as parameters
6      // and initializes the attributes
7
8
9
10
11     // TODO: Create a getter method for width
12
13
14
15     // TODO: Create a getter method for height
16
17
18
19     // TODO: Create a method calculateArea() that returns the area of the
20     // rectangle
21
22
23     // TODO: Create a method calculatePerimeter() that returns the perimeter
24
25
26
27     // TODO: Override the toString() method to return a string in the format:
28     // "Rectangle[width=5.0, height=3.0]"
29
30
31
32 }
```

Task E — Code Debugging

12 P.

The following `BankAccount` class contains several errors. Find and explain all the errors in the code below. Note that errors include both syntax errors that would prevent compilation AND violations of OOP principles (such as encapsulation).

```
1  public class BankAccount {
2      public double balance;
3      private String accountNumber;
4
5      public void BankAccount(String accountNumber, double initialBalance) {
6          accountNumber = accountNumber;
7          balance = initialBalance;
8      }
9
10     public String getAccountNumber() {
11         return this.accountNumber;
12     }
13
14     public deposit(double amount) {
15         balance = balance + amount;
16     }
17
18     public void withdraw(double amount) {
19         balance - amount;
20     }
21
22     public String toString() {
23         "Account: " + accountNumber + ", Balance: " + balance;
24     }
25 }
```

List all errors you found and explain what is wrong:

3. Practical Part: 60min

Task F — Word/Text Analyzer

40 P.

Create a program that analyzes a given text and provides various statistics about it.

Your program should analyze text and provide the following information:

- Count the total number of words
- Count the total number of characters (excluding spaces)
- Find and return the longest word
- Count how many times a specific word appears (case-insensitive)



Tip

You can use the `String.split()` method to split text into words. Example:

```
1 String text = "Hello world from Java";  
2 String[] words = text.split(" ");  
3 System.out.println("Number of words: " + words.length); // 4
```

Java

You can also use `String.toLowerCase()` for case-insensitive comparison and `String.replace()` to remove spaces. For finding the length of a string, use the `length()` method on a string.

1. Create a new project. Give the project a name that contains your student ID number and your name. Create a `TextAnalyzer` class with appropriate methods. 3 P.
2. Create a method `countWords(String text)` that counts and returns the total number of words in the given text. 5 P.
3. Create a method `countCharacters(String text)` that counts and returns the total number of characters, excluding spaces. 5 P.
4. Create a method `findLongestWord(String text)` that finds and returns the longest word in the text. If there are multiple words with the same length, return any one of them. 7 P.
5. Create a method `countWordOccurrences(String text, String word)` that counts how many times a specific word appears in the text. The comparison should be case-insensitive (e.g., "Hello" and "hello" should be counted as the same word). 7 P.
6. Create a main method that demonstrates all the functionality with test cases. Test with at least one sample text that shows all methods working correctly. 5 P.
7. Ensure your methods handle edge cases properly (empty strings, null values, single words, etc.) and validate input appropriately. 5 P.
8. When programming your solution, pay attention to the common coding styles that were established in the lecture (naming conventions, indentation, comments). 3 P.

In total, 105 + 0 pts. are achievable. You have achieved _____ pts. out of 105 pts.

Points	105-94	93-84	83-73	72-63	62-0
Value	very good	good	satisfactory	sufficient	failed